Multiscale Neighborhood Attention Transformer With Optimized Spatial Pattern for Hyperspectral Image Classification

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***Abstract*— Hyperspectral images (HSIs) provide hundreds of continuous spectral bands and have been widely used for fine identification of targets with similar appearances. In earlier studies, convolutional neural networks (CNNs) have been an effective method for HSIs' classification due to their powerful feature extraction capabilities. Recently, self-attention (SA)-based vision transformer (ViT) architecture has been widely explored to fully represent global information. However, most existing transformer-based models primarily focus on global relationships and lack the ability to capture the multiscale features which are crucial for HSIs' classification. This limitation results in inferior performance for transformer-based methods compared with state-of-the-art CNN-based models. To solve this problem, a novel network called multiscale neighborhood attention transformer (MSNAT) is proposed in this article. Unlike previous transformer-based models, MSNAT emphasizes the neighborhood pixels within a local window size and extracts multiscale spatial information using different local window sizes. In addition, a spatial transformation (ST) module is integrated to generate optimized spatial input. The effectiveness of the proposed MSNAT is verified on three real hyperspectral datasets including University of Pavia (UP), University of Houston (UH), and University of Trento (UT). Experimental results demonstrate that the proposed MSNAT method outperforms both CNNs and existing transformer-based models, achieving state-of-the-art classification performance with an overall accuracy of 93.34%, 86.26%, and 96.63% on UP, UH, and UT, respectively. The source code will be available at https://github.com/xinqiao123/MSNAT.**

***Index Terms*— Classification, hyperspectral images (HSIs), multiscale neighborhood attention transformer (MSNAT), spatial transformation (ST) module.**

I. Introduction

With the rapid growth of remote sensing technology, the availability of hyperspectral images (HSIs) has significantly increased. Different from natural scene images which typically consist of three channels (R, G, and B), the HSIs encompass hundreds and continuous spectral bands ranging from visible to infrared [1], [2]. This abundant spectral information allows hyperspectral remote sensing images for fine identification of materials with similar appearance, making HSIs a valuable tool for a wide range of Earth observation (EO) tasks, such as mineral mapping [3], environmental monitoring [4], [5], and precision agriculture [6], [7], [8]. HSIs' classification distinguishes the categorical label for each pixel based on extracted spectral spatial features and attracts significant interest in the remote sensing community [9], [10], [11].

Previous research on classifying HSIs has relied on traditional machine learning algorithms, such as random forest (RF) [12], [13], support vector machine (SVM) [14], [15], [16], and maximum likelihood estimation [17]. However, these methods require manual feature engineering and only extract low-level features, limiting their classification performance. In recent years, deep learning methods, particularly the convolutional neural networks (CNNs), have shown remarkable capabilities in various computer vision tasks, including HSIs' classification. The first study of incorporating deep learning into HSI classification was introduced by Chen et al. [18]. They proposed a hybrid framework that included principle component analysis (PCA), stacked autoencoders, and logistic regression. The stacked autoencoders were used to learn useful high-level features, and the results on a commonly used hyperspectral dataset demonstrated the effectiveness of deep-learning-based models. Since then, CNN has become a popular framework for HSIs' classification. Makantasis et al. [19] designed a deep-learning-based model that used a CNN to hierarchically extract high-level spectral–spatial features and multilayer perceptron (MLP) as the classifier. Zhong et al. [20] developed an end-to-end spectral–spatial residual network (SSRN) that captured spectral–spatial features directly from the raw 3-D HSIs without additional feature engineering. Besides, the residual connection in spectral and spatial blocks effectively solved the problem of decreasing accuracy with a deeper network and achieved state-of-the-art performance. He et al. [21] designed a multiscale 3-D CNN that jointly extracted 2-D multiscale spatial information and 1-D spectral features. Their network was trained via end-to-end modes without any preprocessing. Roy et al. [22] presented a hybrid

Manuscript received 14 August 2023; accepted 8 September 2023. Date of publication 12 September 2023; date of current version 27 September 2023. The work of Weimin Huang was supported in part by the Natural Sciences and Engineering Research Council of Canada Discovery Grants under Grant NSERC RGPIN-2017-04508; and in part by Seed, Bridge and Multidisciplinary Fund from Memorial University of Newfoundland. *(Corresponding author: Weimin Huang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3314550

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spectral CNN (HybridSN) consisting of spectral–spatial 3-D CNN blocks and spatial 2-D CNN modules. The former was used to learn spectral–spatial feature representations at the same time, and the latter was used to extract abstract-level spatial features. Yu et al. [23] constructed a dual-channel CNN (DCNN) that considered the entire image rather than the small patches, to exploit multiscale information. Qiao et al. [24] proposed the use of sharpened cosine similarity (SCS) operation as a replacement for the convolution operation to extract more discriminative features from HSIs and improve the classification performance.

However, the above-mentioned methods did not include the attention mechanisms. Recently, some approaches have leveraged the attention mechanisms to enhance the performance of HSIs' classification. Roy et al. [25] built an attention-based adaptive spectral–spatial kernel improved residual network (A2S2K-ResNet), which extracted discriminative spectral–spatial features through a selective 3-D convolutional kernel and used an efficient channel attention mechanism to further improve classification performance. Hang et al. [26] introduced an attention-aided CNN to explore the discriminative ability of different spectral bands and spatial positions. Such a network included a spectral attention subnetwork and a spatial attention subnetwork to focus on useful bands and locations, respectively. Fang et al. [27] devised a 3-D dense CNN with a spectralwise attention module to capture distinguished spectral feature. Mei et al. [28] proposed a spectral spatial attention network that included two branches: an attention-aided recurrent neural network (RNN) to learn the inner spectral correlation and a CNN with an attention mechanism to extract the relationship among spatial neighborhood pixels. Haut et al. [29] incorporated attention mechanisms into a residual network and introduced an attention-driven model for HSIs' classification. The attention module identified the most representative information from the extracted abstract features, and the experimental results showed competitive advantages of the proposed model. Mou and Zhu [30] designed a spectral attention module to emphasize informative bands and suppress unuseful bands through a gating mechanism. They also demonstrated that the classification performance of CNN could be significantly boosted with the introduction of spectral attention.

Nevertheless, CNN-based models have limitations in global context modeling due to the fixed size of convolutional kernels. To overcome this issue, transformer-based models have been introduced. Transformer is initially a natural language processing model [31] and has been extended to the computer vision area [32], [33]. The self-attention (SA) mechanism can take global information of the input sequence into account. Recently, transformer-based models have been used for HSIs' classification [34], [35]. Zhong et al. [36] developed a spectral–spatial transformer network (SSTN) to overcome the constraints of long-range interaction caused by the convolutional kernel. The implemented network included spatial attention and spectral association modules. In addition, they introduced a factorized architecture search (FAS) framework to determine the optimal order of different blocks. Hong et al. [37] reconsidered HSIs from a spectral sequential perspective and built the SpectralFormer based on the transformer. SpecralFormer could capture the spectral sequence information from original HSIs and achieve significant superiority. He et al. [38] constructed HSI-BERT, a bidirectional encoder representation using transformer to classify HSIs. HSI-BERT provided a global receptive field and captured the global relationship, overcoming the restrictions of limited receptive fields in CNNs. Yu et al. [39] proposed a multilevel spectral–spatial transformer network (MSTNet) to solve the problem of limited receptive field and information loss of downsampling layers in CNN-based methods. The pure transformer encoder was used to extract global features from input sequences with positional embedding. Then, the decoder generated classification results with the same original image size. Sun et al. [40] designed a spectral–spatial feature tokenization transformer (SSFTT) model to learn high-level spectral-spatial semantic features. The SSFTT model used hybrid 2-D CNN and 3-D CNN layers to learn shallow features first. The extracted features were then input into a transformer encoder after Gaussian weighted feature tokenization for high-level feature representation. The final classification results were obtained through a linear layer. Roy et al. [41] put forward a novel multimodal fusion transformer (MFT) network for HSIs' classification. To deal with the problem of random initialization of class tokens, they used light detection and ranging (LIDAR) data as the source of multimodal data to initialize the external class tokens and achieved better generalization capability.

Although the existing transformer-based methods have demonstrated impressive results in HSI classification by modeling global dependencies effectively, they are not fully considered the local and multiscale spatial features, which have been shown to be essential in enhancing classification performance in the computer vision community. For example, Chen et al. [42] addressed the issue of capturing multiscale feature representations in transformer models for image classification. They proposed a dual-branch transformer approach that combined image patches of different sizes to generate multiscale features and fused them using attention mechanisms. However, their method relied on predefined fixed scales for feature extraction. Fan et al. [43] introduced multiscale vision transformers (MViTs) for image recognition. They established connections between multiscale features in hierarchical transformer models. Later, they presented an improved version, MViTv2 [44], which incorporated decomposed relative positional embeddings and residual pooling connections. In addition to the classification tasks, the multiscale features are also used for object detection and semantic segmentation. For example, Zhang et al. [45] constructed an iterative multiscale feature aggregation (IMFA) to fuse multiscale features in the transformer-based object detector. Gu et al. [46] proposed a high-resolution vision transformer (HRViT) to learn multiscale semantic representations.

However, the multiscale transformer-based methods are not widely explored in HSIs' classification tasks [47], [48]. The

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existing methods, such as MCAL [49], extracted multiscale features through embedding convolutional layers with different kernel sizes to the transformer encoder, rather than directly capturing multiscale features through the attention mechanism. In light of this limitation, a novel network called multiscale neighborhood attention transformer (MSNAT) is proposed in this article. The developed MSNAT is also different from the methods used in the computer vision communities where most methods extract multiscale features through the input with various scales or the intermediate feature pyramids. The proposed MSNAT introduces a novel implementation that directly embeds the multiscale features using local attention with different receptive fields. This approach enhances the ability of our model to capture discriminative features across multiscales and facilitates classification performance. The MSNAT is designed to capture neighborhood information at multiple scales in the spatial domain. Furthermore, a spatial transformation (ST) module is integrated into the MSNAT architecture to better handle spatial variability and improve classification performance. The designed MSNAT architecture offers a new perspective to address the limitations of transformer-based models and demonstrates superior performance in HSI classification compared with other state-of-the-art transformer-based approaches. The main contributions of this article can be summarized as follows.

1) A multiscale neighborhood attention (MSNA) module is proposed for HSI classification. This allows tokens to localize attention to their neighborhoods and extract multiscale spatial features effectively, which are often ignored in most existing transformer-based methods.
2) A pointwise convolutional module (PCM), which is rarely used in transformer-based models for HSI classification, is integrated into the transformer encoder here. Compared with the most commonly used MLP, the PCM module is able to more effectively handle the extracted multiscale features while preserving their spatial structure. Besides, a multiscale feature extraction (MSFE) block is further developed based on the proposed MSNA and the PCM.
3) To address the issue of spatial variability, an ST module is integrated into the proposed architecture. Thus, an MSNAT architecture with optimized input features is constructed for HSI classification.
4) The effectiveness of the proposed MSNAT network is evaluated through experiments on three public hyperspectral datasets: University of Pavia (UP), Houston, and Trento. The experimental results demonstrate that MSNAT outperforms other transformer-based methods in terms of various performance metrics, indicating its superiority in HSI classification.

This article is structured as follows. In Section II, a comprehensive explanation of the proposed model, MSNAT, is provided, including its overall architecture and the MSNA mechanism. Section III outlines the experimental setup and evaluates the performance of MSNAT on three public hyperspectral datasets. In addition, the results are discussed, and the advantages of MSNAT are analyzed by comparing it with some representative methods. Finally, Section IV concludes the article with some remarks and suggestions for future work.

II. Methods

Assume that $I_{\text{org}} \in \mathbb{R}^{H\times W\times D}$ represents the original HSI, where H, W, and D denote the spatial height, width, and spectral bands, respectively. Overlapping patches of size P are extracted from the original image and denoted as $X \in \mathbb{R}^{P\times P\times D}$. The label of the central pixel in each patch is considered as the label for the patch. The proposed MSNAT architecture comprises three primary components: convolutional blocks, ST modules, and MSFE blocks, as shown in Fig. 1. The components are described in detail in Sections II-A–II-C, respectively.

A. Convolutional Blocks

To tackle the challenge posed by the high dimensionality of HSIs, convolutional blocks are incorporated in this study to reduce the spectral dimension of HSIs. Given an input patch $X \in \mathbb{R}^{P\times P\times D}$, 2-D convolution operation is first performed using a set of learnable kernels $W \in \mathbb{R}^{k\times k\times C_{\text{in}}\times C_{\text{out}}}$ to generate the output feature $X' \in \mathbb{R}^{P\times P\times C_{\text{out}}}$. k is the kernel size, C_{in} is the number of input channels, which is equal to the number of spectral bands for the first convolutional layer, and C_{out} is the number of output channels. The 2-D convolutional layer followed by a batch normalization (BN) layer could be expressed as follows:

$$X' = \text{BN}(W * X + b) \tag{1}$$

where $*$ represents the convolution operation. W and b denote the weights and biases of the convolutional layer, respectively, and are updated through the gradient descent optimization [50]. The BN layer normalizes the output features to stabilize the training process and improve the performance of the model.

The designed convolutional block effectively leverages the powerful capabilities of CNNs to capture the underlying patterns and structure of HSI data. Meanwhile, the dimensionality reduction on spectral dimension facilitates subsequent processing, such as ST and MSFE blocks.

B. ST Module

Spatial semantic variabilities caused by material mixtures or atmospheric effects present a significant challenge for HSI classification [51]. To solve this issue, an ST module is integrated into our proposed network. The ST module has the capability to learn STs, such as rotation and translation, on the input feature maps. Thus, the developed model can automatically correct the spatial distortions, improving the classification performance. Moreover, the devised ST module can be regarded as data augmentation through STs, alleviating the overfitting problem resulting from the limited number of labeled samples in the HSI dataset.

The ST module, as shown in the left box in Fig. 1, is composed of three main components: the localization,

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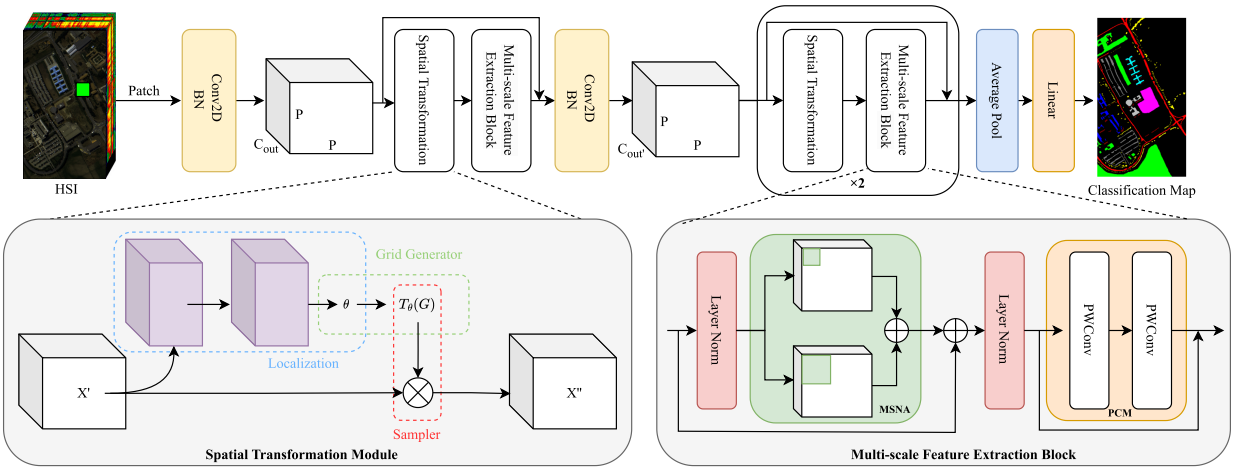


Fig. 1. Overall architecture of the proposed MSNAT network.

grid generator, and sampler. The localization component is responsible for generating the parameters of affine transformation using convolutional and fully connected layers. These predicted transformation parameters are then used to create the sampling grid, which consists of a set of sample points where the input feature maps are sampled to produce the transformed output. Finally, the sampler component generates the transformed output feature maps based on the sampling grid and input feature maps. The spatial optimized feature maps can be generated as follows:

$$X'' = f_{\mathrm{ST}}(X') \tag{2}$$

where X' and X'' represent the input and output feature maps of the ST module, respectively. f_{ST} defines the operation of the ST module and is described in detail below.

1) Localization: To be specific, the ST module takes the output feature maps of the convolutional block $X' \in \mathbb{R}^{P\times P\times C_{\mathrm{out}}}$ as input. X' is first fed into the localization component to generate the parameters θ of the affine transformation. Table I describes the architecture of our designed localization component, which consists of two convolutional layers followed by max pooling layers and ReLU activation function, and additional two fully connected layers. The localization component predicts the parameters of transformation according to the following equation:

$$\theta = f_{\mathrm{loc}}(X') = \begin{bmatrix}\theta_{11}, & \theta_{12}, & \theta_{13}\\ \theta_{21}, & \theta_{22}, & \theta_{23}\end{bmatrix} \tag{3}$$

where θ consists of six values. Among these, $\theta_{11}, \theta_{12}, \theta_{21}, \theta_{22}$ determine the angle of rotation, and θ_{13}, θ_{23} correspond to the translation factor along the width and height of the input feature maps, respectively. All those parameters are obtained from the output of the linear layer in the localization module and are updated iteratively during the model training process.

2) Grid Generator: The grid generator takes the predicted transformation parameters θ and the target coordinates in the output feature maps X'' as input. It then produces corresponding source coordinates in the input feature maps X'. These source coordinates are subsequently used to define sample points and form the sampling grid. Grid generator

TABLE I
ARCHITECTURE OF LOCALIZATION COMPONENT IN ST MODULE

Layer	Parameters	Stride	Padding
Convolutional	16, 5 × 5	1	0
Max pooling	2 × 2	1	0
Convolutional	16, 5 × 5	1	0
Max pooling	2 × 2	1	0
Linear1	32	/	/
Linear2	6	/	/

$T_\theta(G)$ could be written as

$$\begin{bmatrix}x_i^s\\ y_i^s\end{bmatrix} = T_\theta(G) = \begin{bmatrix}\theta_{11}, & \theta_{12}, & \theta_{13}\\ \theta_{21}, & \theta_{22}, & \theta_{23}\end{bmatrix}\begin{bmatrix}x_i^t\\ y_i^t\\ 1\end{bmatrix} \tag{4}$$

where $(x_i^s, y_i^s)^T$ are the source coordinates, and $(x_i^t, y_i^t)^T$ denote the target coordinates.

3) Sampler: The sampler is the final component of the ST module and is responsible for generating transformed output feature maps X'' using the input feature maps X' and the sampling grid $(x_i^s, y_i^s)^T$ generated by the grid generator. The sampler obtains the value at each sample point via bilinear interpolation, which is a method for estimating the value at a given position using nearby points. Specifically, the value ${X''}_i^c$ at position (x_i^t, y_i^t) in channel c of output feature maps X'' could be calculated as follows:

$${X''}_i^c = \sum_{p_h}^{P}\sum_{p_w}^{P} {X'}_{p_h p_w}^{c} \max\left(0, 1 - \left|x_i^s - p_w\right|\right)\max \times \left(0, 1 - \left|y_i^s - p_h\right|\right) \tag{5}$$

where ${X'}_{p_h p_w}^{c}$ is the value at location (p_h, p_w) in channel c of input feature maps X'. By observing (5), one can find that the value ${X''}^c i$ is calculated as the sum of the product of ${X'}^c p_h p_w$ and their corresponding weights given by the function max().

C. MSFE Block

While transformers have demonstrated great advantages in capturing global long-range relationships for HSI classification, they tend to overlook the importance of multiscale

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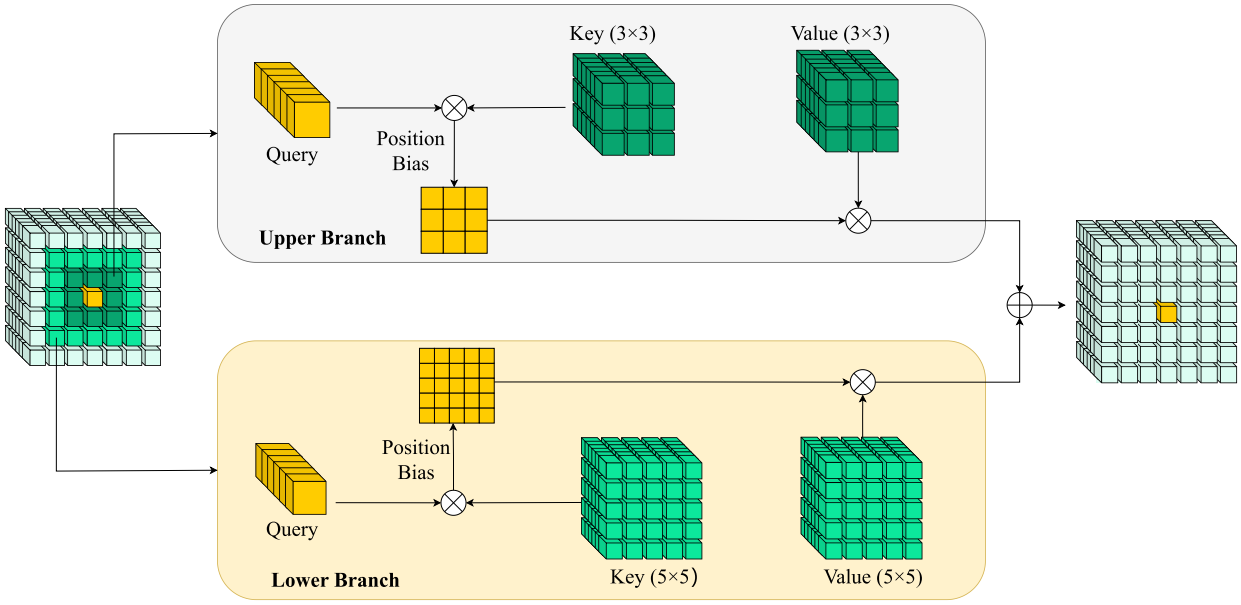


Fig. 2. Multiscale neighborhood attention.

spatial features. To address this issue, an MSNA mechanism is proposed to effectively extract multiscale spatial features. Based on this, an MSFE block is further developed as depicted in Fig. 1. The MSFE block consists of an MSNA module and pointwise convolutional module (PCM). Layer normalization (LN) is applied before MSNA and PCM to standardize the distribution of the input features and improve the training efficiency. The structure of the proposed MSNA and PCM is described in detail below.

1) Multiscale Neighborhood Attention: The MSNA module consists of two parallel branches, each of which is responsible for capturing spatial features at different scales, as illustrated in Fig. 2.

The upper branch of the MSNA mechanism is specifically designed to capture features at a small scale. It first projects the input into query ($Q \in \mathbb{R}^{N\times C}$), key ($K \in \mathbb{R}^{N\times C}$), and value ($V \in \mathbb{R}^{N\times C}$) matrices through 2-D convolutional layers, where N and C represent the numbers of the embedded tokens and spectral dimension, respectively. To capture more diverse and informative features, the branch further divides Q, K, and V into h heads along the spectral dimension, producing h groups of query, key, and value denoted as

$$Q = \left\{ Q_i \in \mathbb{R}^{N\times d} \right\}_{i=1}^{h} \tag{6a}$$

$$K = \left\{ K_i \in \mathbb{R}^{N\times d} \right\}_{i=1}^{h} \tag{6b}$$

$$V = \left\{ V_i \in \mathbb{R}^{N\times d} \right\}_{i=1}^{h} \tag{6c}$$

where d is the dimension for each head and is equal to (C/h).

Next, the neighborhood attention between the query and key is calculated with multiheads, and the attention computation process for each head is performed independently. In the upper branch of the proposed MSNA, the receptive field of each token in the query Q_i is limited to a small local window size, as illustrated in Fig. 2. Besides, the relative position bias B is added to the attention score to retain the position information. Specifically, for a given query token $Q_{i,\mu}$ in the ith head, the neighborhood attention $A^k_{i,\mu}$ within a local window of size k could be obtained as

$$A^k_{i,\mu} = \begin{bmatrix} Q_{i,\mu}K^T_{i,\mu,1} + B_{i,\mu,1} \\ Q_{i,\mu}K^T_{i,\mu,2} + B_{i,\mu,2} \\ \vdots \\ Q_{i,\mu}K^T_{i,\mu,k} + B_{i,\mu,k} \end{bmatrix} \tag{7}$$

where $A^k_{i,\mu}$ is the neighborhood attention of the μth query token within a local window size of k. $K_{i,\mu,j}$ and $B_{i,\mu,j}$ $(j = 1, 2, \ldots, k)$ refer to the jth neighboring key and relative position bias of the μth query token, respectively.

Then, the Softmax function is applied to normalize the obtained attention score. The output of the neighborhood attention for $Q^k_{i,\mu}$ is the weighted sum of the neighborhood values within its local window. The formula is defined as follows:

$$\text{head}_{i,\mu} = \text{softmax}\left(\frac{A^k_{i,\mu}}{\sqrt{d}}\right) V^k_{i,\mu} \tag{8}$$

$$V^k_{i,\mu} = \left[V_{i,\mu,1}, V_{i,\mu,2}, \ldots, V_{i,\mu,k}\right] \tag{9}$$

where d is the dimension of K_i, as defined in (6b). $\sqrt{d}$ is the scaling factor to stabilize the gradient during back propagation. $V^k_{i,\mu}$ represents the values within the local window size of k and $\text{head}_{i,\mu}$ is the output of the μth query token in the ith head. After calculating the neighborhood attention for all the tokens in query Q_i, the output of the ith head could be obtained and is denoted as head_i. The outputs of h heads are then concatenated along the spectral dimension and projected to a final output as

$$Y_1 = (\text{head}_1 \odot \text{head}_2 \odot \cdots \odot \text{head}_h) W^O \tag{10}$$

where $\odot$ represents the concatenation operation along the spectral dimension, and W^O denotes the output projection matrix. Y_1 is the output of the upper branch in the proposed MSNA module.

The lower branch of the MSNA is similar to the upper branch but designed to capture large-scale spatial features.

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Unlike the upper branch, the lower branch uses a larger local window size during the calculation of neighborhood attention. As a result, it is able to extract long-range information from the input data.

Finally, the MSNA module fuses the multiscale features by performing an elementwise addition of the outputs from both the branches. This fusion combines the features from different scales, generating a more comprehensive representation of the input data.

2) Pointwise Convolutional Module: Typical transformer-based models for HSI classification use multilayer perceptron (MLP) in the transformer block to process the input features. MLP consists of two linear layers and an activation function. However, the extracted features from the MSNA module are represented in a 3-D form instead of 2-D form, making it unsuitable to directly input to the linear layers. To handle this problem, a pointwise convolution module (PCM) which is composed of two pointwise convolutional layers is proposed in this study. The PCM module can efficiently capture spatial correlations between different channels of input features while maintaining their spatial structure.

The PCM involves two pointwise convolutional layers which are implemented using 1×1.2-D convolutional filters. Besides, the Gaussian error linear unit (GELU) [52] activation function is incorporated into this module. Specifically, this step can be represented as

$$Z = \mathrm{PCM}\left(\widehat{Y}\right) = \mathrm{PWConv}_2\left(\mathrm{GELU}\left(\mathrm{PWConv}_1\left(\widehat{Y}\right)\right)\right) \quad (11)$$

where $\widehat{Y}$ is the result of MSNA output after LN, and Z is the output of PCM.

To summarize, our proposed MSFE block incorporates an MSNA module and pointwise convolutional layers to extract multiscale spatial features. To prevent overfitting, we also introduce LN and residual connections. The overall computation for this block can be expressed as follows:

$$Y = \mathrm{MSNA}\left(\mathrm{LN}\left(X''\right)\right) + X'' \quad (12)$$

$$Z = \mathrm{PCM}(\mathrm{LN}(Y)) + Y \quad (13)$$

where X'' is the output of the SP module which serves as the input to the MSFE block. Y and Z denote the output of the MSNA module and the FFN module, respectively.

D. Multiscale Neighborhood Attention Transformer

An MSNAT network is developed based on the above-described blocks. The overall architecture of the network is given in Fig. 1. First, small disjointed patches with the size of $11 \times 11 \times$ bands are extracted from the original HSI and are divided into training data, validation data, and test data. Training datasets are used to train and optimize the model weight parameters. Validation data are used to monitor the loss change during the training process and avoid the overfitting problem. The model performance is evaluated on test data finally.

The extracted patches are first input into the convolutional block, which generates a feature map of $11 \times 11 \times 49$ using a 2-D convolutional layer. The selection of 49 channels in the model is determined by considering the balance between reducing the computational cost and preserving spectral information. As in [36], 49 is also selected here. The ST module is then used to optimize the extracted spatial features. The MSFE block, which consists of MSNA and PCM, takes the optimized features as input to capture multiscale spatial features. To maintain the original features, a residual connection is applied between the input of the ST module and the output of the MSFE block. The output of the MSFE block is further processed by a 2-D convolutional layer with 16 channels to reduce the redundant spectral information. Subsequently, the ST module and the MSFE block are repeated twice to capture more complex multiscale features and improve the classification performance.

Finally, the output features with a shape of $11 \times 11 \times 16$ are passed through average pooling and fed into a fully connected layer, followed by a Softmax layer to generate the final classification results.

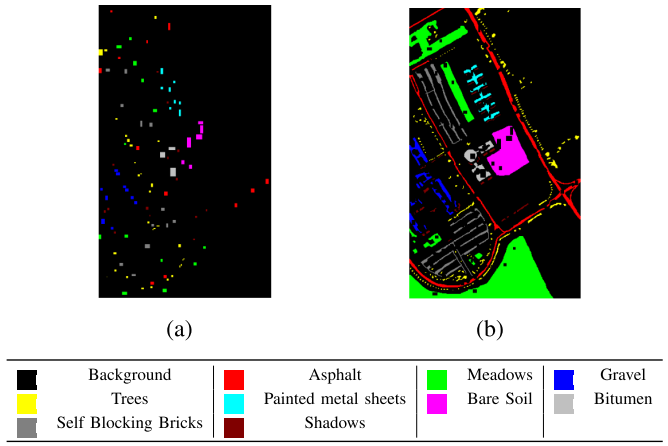


Fig. 3. Spatial distribution of the UP dataset. (a) Training data. (b) Test data.

III. Experiments

In this section, the proposed MSNAT is evaluated on three widely used real-world hyperspectral datasets. In addition to the commonly used UP dataset, two relatively new datasets, the University of Houston (UH) and the University of Trento (UT), are also selected. First, the three datasets used in this study are described concisely. Next, the model configuration and experimental setups are presented. Finally, the classification results are reported, followed by corresponding discussion.

A. Dataset Description

1) University of Pavia: The UP dataset, captured by the Reflective Optics System Imaging Spectrometer (ROSIS) over the UP campus in Northern Italy, is used in this study. The dataset encompasses 610 by 340 pixels with a spatial resolution of 1.3 meters per pixel (MPP). It comprises 103 spectral bands within the range of 430–860 nm. The dataset consists of nine land-cover classes, and all the labeled samples are split into training data and test data, as displayed in Fig. 3.

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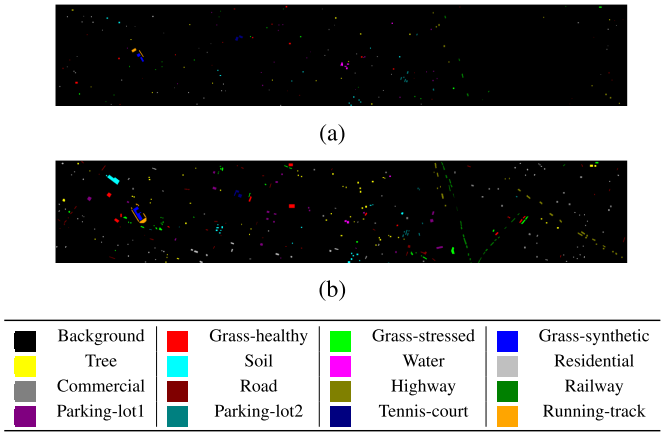


Fig. 4. Spatial distribution of the UH dataset. (a) Training data. (b) Test data.

TABLE II
NUMBER AND RATIO OF TRAINING DATA AND TEST DATA

Datasets	Training data		Test data	
	Number	Percentage	Number	Percentage
UP	3921	8.93%	40002	91.07%
UH	2832	18.84%	12197	81.16%
UT	819	2.71%	29395	97.29%

2) University of Houston: The UH dataset was acquired through Compact Airborne Spectrographic Imager (CASI) and provided by IEEE Geoscience and Remote Sensing Society (GRSS) for the data fusion contest in 2013. The dataset comprises 340 × 1905 pixels with a spatial resolution of 2.5 MPP and contains 144 bands covering a wavelength range from 380 to 1050 nm. There are 15 land-cover classes, and the detailed spatial distributions of the training data and test data are exhibited in Fig. 4.

3) University of Trento: The UT dataset was collected over the south of Trento, Italy. It includes 63 spectral bands ranging from 420 to 990 nm. The dataset contains 600 × 166 pixels with a spatial resolution of 1 MPP. It comprises six land-cover classes, and the disjointed distributions of the training data and test data are illustrated in Fig. 5.

Compared with other datasets such as the Indian Pine (IP), the selected datasets, UH and UT, are new. The number and the ratio of training data and test data are illustrated in Table II. Different from the commonly used random sampling, this study uses a disjointed sampling method. In the HSIs' classification task, patch extraction is performed to simultaneously capture spectral–spatial information, leading to possible spatial overlap. This concern becomes especially significant in the case of random sampling. To address this challenge, the disjointed sampling method is adopted. By constraining the training data to some small areas, this approach effectively mitigates the issue of spatial overlap.

B. Setup

To assess the effectiveness of the proposed MSNAT, eight reference methods are selected for comparison, including three CNN-based networks and five transformer-based networks.

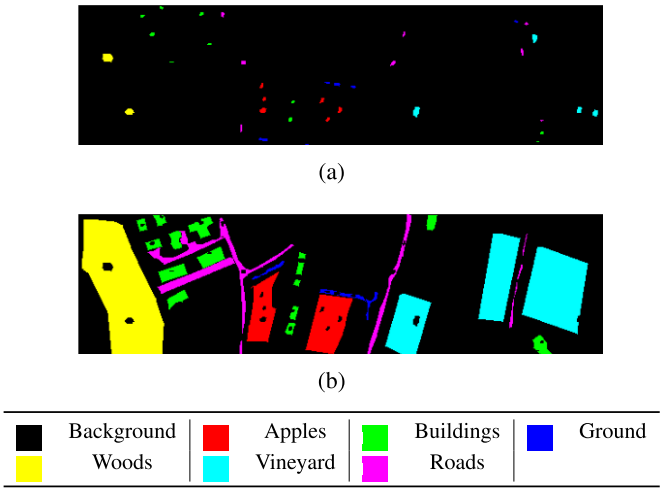


Fig. 5. Spatial distribution of the UT dataset. (a) Training data. (b) Test data.

1) CNN-Based Networks: The proposed MSNAT for HSI classification is compared with three CNN-based approaches, i.e., SSRN [20], A2S2K-ResNet [25], and spectral–spatial ConvNeXt network (SS-ConvNeXt) [53]. The SSRN approach involves four iterative spectral and spatial residual blocks. Each block contains 3-D convolutional layers, 3-D BN, and ReLU activation function. The receptive fields of the 3-D convolutional layers in the spectral and spatial blocks are $(7, 1, 1)$ and $(1, 3, 3)$, respectively. For A2S2K-ResNet, the patch size is consistently preset as 11×11. Furthermore, the parameters within the selective kernel attention module and spectral spatial residual blocks remain unchanged from those presented in the original paper. SS-ConvNeXt is one of the most state-of-the-art models published in 2023. A spectral-ConvNext and a spatial-ConvNeXt block are designed to capture spectral and spatial information, respectively.

2) Transformer-Based Networks: Five transformer-based architectures, including ViT [32], SSTN [36], SSFTT [40], SpectralFormer [37], and MFT [41], are selected for comparison. ViT is the first model to apply the transformer architecture to the field of computer vision. The network architectures of SSTN, SSFTT, and SpectralFormer are kept the same as their respective original papers. However, for MFT, the Lidar data branch used to generate the external class token in the original paper is discarded, and only the hyperspectral data branch is retained for a fair comparison with other methods.

All these models are developed and trained using the PyTorch deep learning framework. Cross-entropy is adopted for the loss calculation, and the Adam optimizer is used to optimize the parameters of the network. The validation samples are a portion of the training data. For the results shown in all the tables, in the training data, 90% are used for training, and the remaining 10% are used for validation for each epoch. In later experiments for evaluating the performance with a small size of training samples, i.e., 3%, 5%, 7%, and 9%, the validation data percentages are 97%, 95%, 93%, and 91%, respectively. The total number of training epochs is set to 500. Early stop strategy is implemented to prevent

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TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DISJOINTED UP DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	CNN-based Methods			Transformer-based Methods					
	SSRN	A2S2K-ResNet	SS-ConvNeXt	ViT	SSTN	SSFTT	SpectralFormer	MFT	MSNAT
1	**97.11 ± 1.63**	93.78 ± 3.77	94.64 ± 1.34	82.30 ± 1.17	92.63 ± 2.51	84.74 ± 1.59	70.69 ± 3.94	82.38 ± 2.60	95.46 ± 1.10
2	71.22 ± 0.63	76.47 ± 7.37	**96.31 ± 0.73**	88.39 ± 1.15	81.88 ± 2.49	94.57 ± 1.44	83.22 ± 1.53	90.98 ± 3.83	93.46 ± 1.51
3	83.43 ± 6.48	71.96 ± 5.99	**90.88 ± 1.08**	51.68 ± 4.96	83.68 ± 8.12	70.34 ± 2.73	48.17 ± 3.42	80.54 ± 9.23	57.26 ± 4.14
4	83.78 ± 5.53	97.15 ± 2.08	94.90 ± 0.87	96.50 ± 0.39	91.59 ± 2.91	97.42 ± 0.96	94.32 ± 0.53	94.13 ± 1.22	**97.26 ± 1.35**
5	**99.53 ± 0.40**	99.37 ± 0.27	98.13 ± 0.33	98.89 ± 0.56	98.78 ± 0.64	99.10 ± 0.64	99.51 ± 0.33	98.76 ± 0.73	98.98 ± 0.90
6	**99.93 ± 0.11**	97.73 ± 2.23	70.23 ± 6.02	48.43 ± 2.57	95.25 ± 5.50	88.77 ± 2.11	57.75 ± 1.81	96.75 ± 1.83	94.86 ± 5.34
7	**99.88 ± 0.12**	97.61 ± 0.86	98.43 ± 1.81	72.74 ± 4.33	97.78 ± 1.90	85.65 ± 7.07	59.14 ± 3.03	97.59 ± 2.38	98.23 ± 1.46
8	98.34 ± 1.20	**99.54 ± 0.32**	97.78 ± 0.41	96.11 ± 0.68	96.15 ± 0.99	97.21 ± 0.79	91.93 ± 0.39	97.84 ± 0.50	99.10 ± 0.55
9	**99.22 ± 0.52**	97.66 ± 0.55	97.08 ± 0.51	94.04 ± 2.82	93.99 ± 1.25	97.46 ± 0.53	94.47 ± 1.93	98.04 ± 0.70	94.89 ± 2.53
OA	84.38 ± 0.58	86.44 ± 3.12	92.96 ± 0.56	82.46 ± 0.48	88.19 ± 1.67	91.65 ± 0.66	78.37 ± 0.76	91.13 ± 1.77	**93.34 ± 1.02**
AA	92.49 ± 0.88	92.36 ± 1.05	**93.15 ± 0.88**	81.01 ± 0.63	92.41 ± 1.41	90.58 ± 0.76	77.69 ± 0.67	93.00 ± 1.26	92.16 ± 1.23
Kappa	80.06 ± 0.73	82.60 ± 3.83	90.42 ± 0.79	76.32 ± 0.62	84.62 ± 2.12	88.81 ± 0.86	71.33 ± 0.91	88.27 ± 2.25	**91.11 ± 1.36**

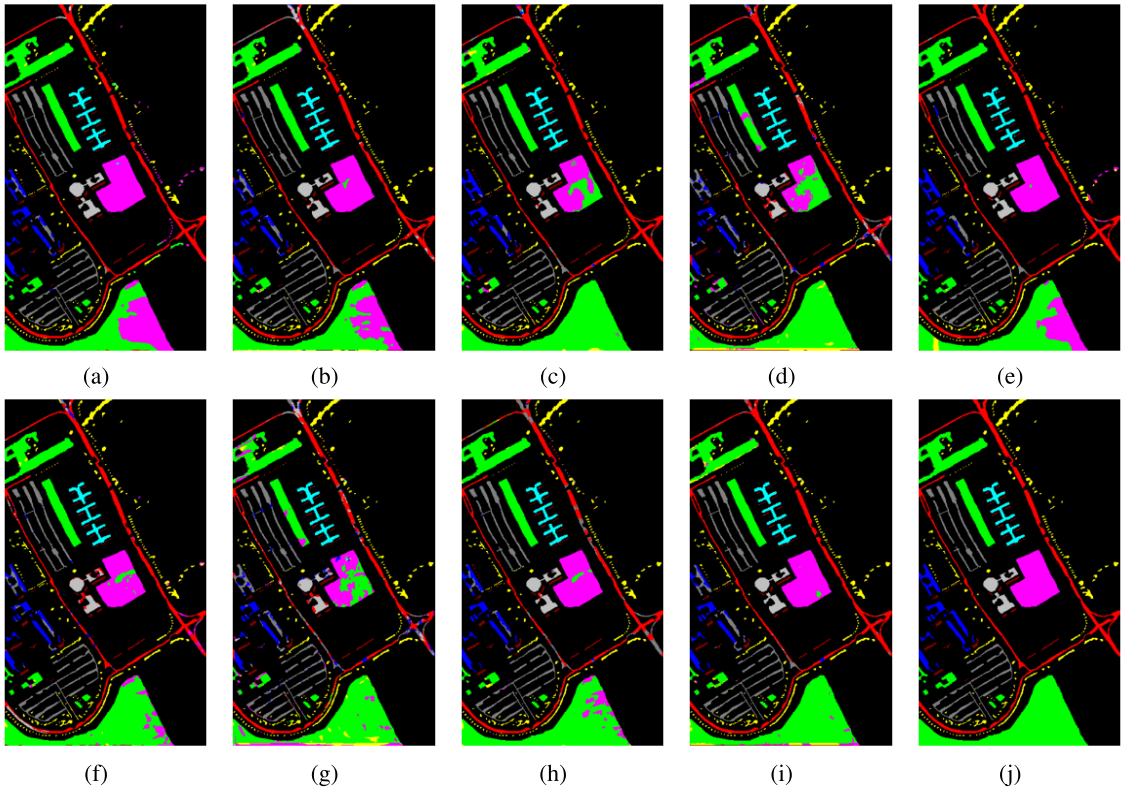


Fig. 6. Classification maps generated by different models on the disjointed UP dataset. (a) SSRN. (b) A2S2K-ResNet. (c) ConvNeXt. (d) ViT. (e) SSTN. (f) SSFTT. (g) SpectralFormer. (h) MFT. (i) MSNAT. (j) Ground truth.

overfitting and reduce training time. Specifically, if the loss on the validation dataset does not decrease after 50 consecutive epochs, the training process stops and the weights with the minimum validation loss are saved for further test. Overall accuracy (OA), average accuracy (AA), and Kappa coefficient are calculated as the classification metrics. To evaluate the generalization capability of different approaches, each model is run five times and the mean values and standard deviations are reported. All the experiments are conducted using a GPU V100 with 32 GB of RAM on a high-performance computing (HPC) server provided by Compute Canada.

Given the high dimensionality and complexity of HSIs, data preprocessing plays a critical role in the classification process. In this study, the min-max scaler approach is used to normalize the spectral dimension of the data into a range of [0, 1]. This preprocessing allows us to achieve stable training and optimize the performance of the models.

C. Comparison With State-of-the-Art Methods

1) Quantitative Analysis: The classification results of different methods on the three datasets are displayed in Tables III–V. The best results are highlighted in bold. The study has demonstrated that the proposed MSNAT outperforms other CNN and transformer-based models, thereby exhibiting superior classification performance.

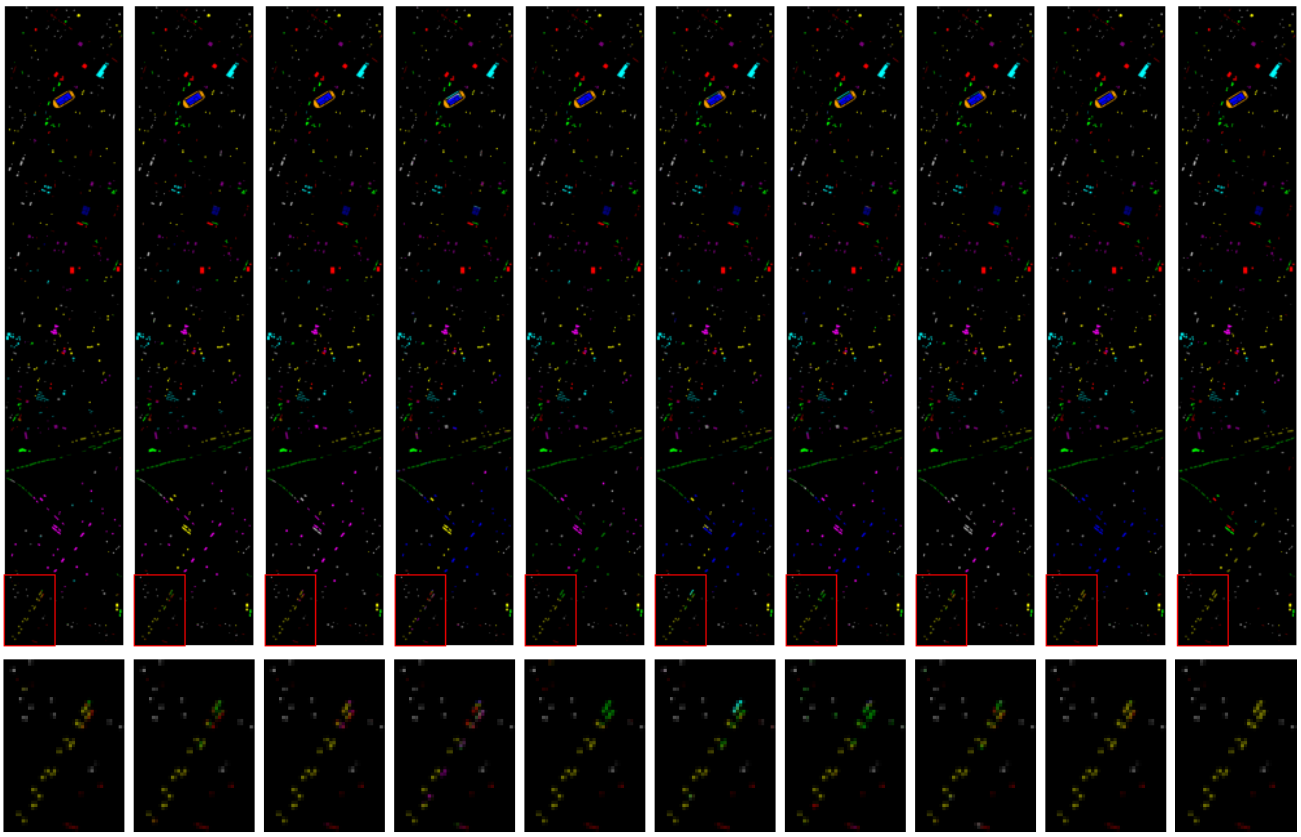
Table III lists the classification results of various models for the UP dataset. As shown in this table, SS-ConvNeXt achieves the second-best accuracies in terms of OA and Kappa, demonstrating its strong feature representation capability. Another two CNN-based models, i.e., SSRN and A2S2K-ResNet, exhibit a similar classification performance. However, they are outperformed by most transformer-based models, such as SSTN and SSFTT, indicating the superiority of transformer-based models in HSI classification. Among the

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TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DISJOINTED UH DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	CNN-based Methods			Transformer-based Methods					
	SSRN	A2S2K-ResNet	SS-ConvNeXt	ViT	SSTN	SSFTT	SpectralFormer	MFT	MSNAT
1	82.30 ± 0.40	82.28 ± 0.21	82.09 ± 0.22	80.97 ± 0.81	83.34 ± 3.24	82.60 ± 0.40	81.16 ± 0.60	81.33 ± 0.68	**82.83 ± 0.31**
2	**85.13 ± 0.04**	84.87 ± 0.18	85.09 ± 0.08	83.93 ± 0.51	80.45 ± 6.07	84.47 ± 0.62	83.31 ± 1.27	83.97 ± 1.05	84.72 ± 0.47
3	96.83 ± 2.31	96.71 ± 0.74	**99.17 ± 0.15**	61.15 ± 11.96	98.57 ± 1.14	87.60 ± 7.68	80.55 ± 4.15	96.63 ± 2.23	98.50 ± 0.94
4	92.99 ± 0.33	95.87 ± 3.30	91.25 ± 1.68	88.60 ± 2.11	86.16 ± 3.99	**96.69 ± 3.41**	85.93 ± 1.48	95.08 ± 3.10	92.65 ± 0.37
5	**100.0 ± 0.00**	**100.0 ± 0.00**	99.98 ± 0.04	97.12 ± 0.48	99.02 ± 0.80	99.89 ± 0.07	96.82 ± 1.05	99.41 ± 0.16	99.87 ± 0.18
6	95.94 ± 0.28	95.94 ± 0.28	**99.72 ± 0.34**	92.17 ± 3.44	95.80 ± 0.00	87.41 ± 4.92	87.69 ± 6.97	95.80 ± 3.19	96.64 ± 2.98
7	85.19 ± 2.19	**87.74 ± 1.45**	86.98 ± 1.76	85.69 ± 0.94	81.81 ± 6.56	83.68 ± 2.76	77.18 ± 3.60	86.12 ± 5.59	86.34 ± 2.86
8	66.13 ± 5.87	70.31 ± 1.72	70.24 ± 3.93	63.17 ± 4.48	**75.44 ± 2.41**	70.75 ± 3.19	56.43 ± 6.52	74.19 ± 2.13	71.93 ± 3.41
9	76.60 ± 2.15	81.95 ± 3.90	82.49 ± 2.16	79.57 ± 4.08	79.26 ± 12.99	84.76 ± 1.66	75.73 ± 3.25	82.66 ± 1.57	**85.93 ± 1.39**
10	59.23 ± 5.59	50.10 ± 1.72	53.49 ± 5.08	43.19 ± 3.53	56.54 ± 6.69	51.24 ± 3.50	44.61 ± 2.06	56.64 ± 6.67	**64.92 ± 1.58**
11	81.73 ± 3.11	80.11 ± 1.17	83.28 ± 1.50	68.86 ± 3.05	**88.71 ± 9.50**	77.86 ± 2.52	61.44 ± 3.11	79.13 ± 2.45	78.98 ± 2.44
12	92.37 ± 5.80	90.62 ± 4.42	86.57 ± 2.05	73.66 ± 0.57	91.39 ± 7.53	86.92 ± 2.04	82.02 ± 1.32	89.57 ± 5.02	**97.27 ± 0.77**
13	90.39 ± 3.14	**92.70 ± 0.34**	87.72 ± 2.22	85.54 ± 1.62	70.39 ± 4.75	89.96 ± 1.43	63.58 ± 3.32	83.02 ± 3.05	91.51 ± 1.07
14	**100.0 ± 0.00**	97.89 ± 1.29	**100.0 ± 0.00**	92.47 ± 6.06	99.11 ± 1.21	97.65 ± 4.10	83.89 ± 2.32	98.06 ± 2.39	99.76 ± 0.32
15	99.24 ± 0.61	98.69 ± 1.52	**100.0 ± 0.00**	81.61 ± 7.29	96.41 ± 5.24	88.16 ± 9.62	70.36 ± 2.47	92.05 ± 6.17	97.42 ± 2.86
OA	84.18 ± 1.12	84.38 ± 0.71	84.29 ± 0.54	76.84 ± 0.17	83.69 ± 2.44	82.99 ± 1.02	74.70 ± 1.07	84.25 ± 1.16	**86.26 ± 0.47**
AA	86.94 ± 0.93	87.05 ± 0.52	87.21 ± 0.45	78.51 ± 0.44	85.49 ± 1.79	84.64 ± 1.08	75.38 ± 0.34	86.24 ± 0.85	**88.62 ± 0.52**
Kappa	82.96 ± 1.21	83.15 ± 0.74	83.06 ± 0.59	74.97 ± 0.18	82.33 ± 2.61	81.61 ± 1.10	72.69 ± 1.13	82.97 ± 1.23	**85.16 ± 0.50**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. Classification maps generated by different models on the disjointed UH dataset. Some regions where the MSNAT method yields better classification maps have been highlighted within the red box and zoomed in below. (a) SSRN. (b) A2S2K-ResNet. (c) SS-ConvNeXt. (d) ViT. (e) SSTN. (f) SSFTT. (g) SpectralFormer. (h) MFT. (i) MSNAT. (j) Ground truth.

transformer-based models, the proposed MSNAT achieves the highest average OA of 93.34% and outperforms the second-best transformer-based model by approximately 2%. This improvement could be attributed to the effectiveness of the proposed model in capturing multiscale features for transformers.

Similar classification results can also be found in the UH dataset, as illustrated in Table IV. It can be observed that both the ViT and SpectralFormer methods achieve relatively lower average OA of 76.84% and 74.70%, respectively. The proposed MSNAT achieves higher accuracy with OA, AA, and Kappa values being 86.26%, 88.62%, and 85.16%, outperforming other state-of-the-art HSI classification methods. The superior performance of the proposed MSNAT model illustrates the designed MSFE block could effectively extract both local and multi-scale features, enhancing the feature representation capability of transformer-based models. Based on this, it can be concluded that

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incorporating multiscale features is crucial for achieving high accuracy.

Table V displays the classification results obtained on the UT dataset. It could be found that the SS-ConvNeXt model showcases lower accuracies possibly due to its higher complexity compared with the SSRN and A2S2K-ResNet models, causing the overfitting problem on the UT dataset. Besides, another two CNN-based methods achieve better classification results compared with the transformer-based methods. This may be due to the higher spatial resolution of the UT dataset and CNN-based models are more suitable for local and fine-grained spatial feature extraction, resulting in better classification performance on the UT dataset. In contrast, the existing transformer-based methods focus on capturing long-range dependencies but overlook the detailed local information. The proposed MSNAT model overcomes this limitation by incorporating local and MSFE mechanisms, leading to improved classification performance. Specifically, the MSNAT model achieves an average OA, AA, and Kappa of 96.63%, 94.12%, and 95.49%, outperforming the MFT and SSRN models by 2.86% and 0.92%, respectively.

2) Qualitative Visual Comparison: Figs. 6–8 display the classification maps generated by the aforementioned methods for the UP, UH, and UT datasets, respectively. These classification maps provide a qualitative visual comparison of different methods. To enhance the visualization of the UH dataset, some regions where the MSNAT method yields better classification maps have been highlighted within the red box and zoomed in below. As shown in Figs. 6–8, the ViT model produces more noisy and less accurate classification maps since it solely depends on global SA mechanisms to extract relationships in a long range, ignoring the incorporation of local spatial information. On the other hand, the classification maps generated by the proposed MSNAT method have less noise and are closer to the ground truth when compared with other methods, demonstrating its advantage of MSFE.

To further evaluate the effectiveness of the proposed MSNAT method, t-distributed stochastic neighbor embedding (t-SNE) is used to depict the extracted features from the UP, UH, and UT datasets. Figs. 9–11 display the feature distributions of different methods on the selected datasets. For the proposed MSNAT, it can be observed that there is some mixing between certain classes. For example, in Fig. 11(i) for the UT dataset, class 2 (Buildings) and class 6 (Roads) overlap, which is also reflected in the misclassified pixels in the classification map shown in Fig. 8(i). However, the overall t-SNE visualization demonstrates that samples belonging to the same categories are clustered together, while different classes exhibit significant interclass variance and are well-differentiated. Take class 6 (Bare Soil) in Fig. 9 as an example, the distribution produced by the proposed method is tightly concentrated while those generated by other methods exhibit mixing to some extent [see Fig. 9(c) and (d)]. This observation further supports the superior classification performance of the MSNAT model.

3) Computational Cost Comparison: Table VI presents the parameter size and inference time of various methods evaluated on the test data of UP, UH, and UT. The inference time is calculated on the 12th Gen Intel[1] Core[2] i7-12700F at 2.10 GHz with 16 GB of RAM and the RTX 3070 GPU. It could be observed that the SSTN has the least number of parameters and inference time. However, its classification accuracy is relatively lower. The proposed MSNAT method could achieve better performance with a reasonable inference time.

D. Robustness Assessment

To assess the robustness of various models, experiments are conducted considering different numbers of training samples. Specifically, small subsets that are randomly selected 1%, 3%, 5%, 7%, and 9% of the original training data are used for training, and the test data remain unchanged. Fig. 12 displays the overall accuracies of both the CNN and transformer-based models on the UP, UH, and UT datasets for varying numbers of training data, and the proposed MSNAT is labeled by the light green curves. The accuracy results achieved with 1% training samples are zoomed in for better visualization. The higher accuracies reveal the superior performance of the MSNAT model Besides, for the UH dataset, the proposed MSNAT model outperforms other models even with a limited number of training samples, as shown in Fig. 12(b). Thus, it can also be concluded that the proposed MSNAT model has strong robustness with respect to various sizes of training data.

E. Ablation Study

1) Impact of ST Module: To investigate the impact of the ST module on the proposed MSNAT model, the models with and without the ST module are compared, and the results are displayed in Table VII, where symbol ✓indicates with the module. The table shows that the ST module improves the classification performance of the MSNAT on the UT dataset in terms of OA, AA, and Kappa. This finding indicates the effectiveness of the ST module in the MSNAT model. However, for the UP and UH datasets, the model with the ST module exhibits slightly lower accuracies. This observation could potentially be attributed to the spatial resolution of the datasets. A lower resolution might introduce imprecise geometric variations, making it more challenging to generate accurate transformations.

2) Impact of MSFE Block: To explore the impact of the MSFE block, experiments with and without MSFE block are conducted. The classification results from the three different datasets are presented in Table VII. As the table shows, when the MSFE block is included, the model achieves higher classification accuracies, highlighting the benefit of the MSFE block in enhancing the classification performance of the MSNAT.

3) Impact of Pointwise Convolution: To demonstrate the effectiveness of pointwise convolutions, comparative experiments with original MLP are carried out, and the results are shown in Table VIII. It could be found that using the PCM obtains higher classification accuracies, illustrating its effectiveness in preserving the original structure.

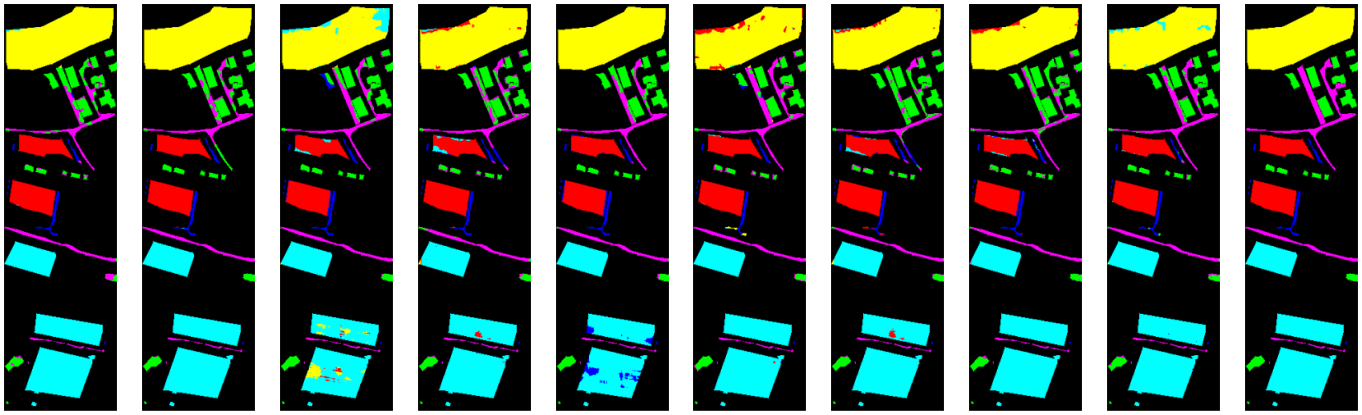
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TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DISJOINTED UT DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	CNN-based Methods			Transformer-based Methods					
	SSRN	A2S2K-ResNet	SS-ConvNeXt	ViT	SSTN	SSFTT	SpectralFormer	MFT	MSNAT
1	**99.31 ± 0.29**	99.07 ± 0.41	94.30 ± 0.78	95.22 ± 1.35	94.80 ± 1.69	98.54 ± 0.44	96.34 ± 0.27	95.58 ± 0.92	98.29 ± 0.73
2	94.87 ± 3.03	**97.21 ± 0.71**	91.78 ± 0.98	90.78 ± 0.62	95.00 ± 3.74	87.38 ± 3.35	91.93 ± 1.59	92.04 ± 3.33	96.13 ± 1.54
3	98.45 ± 1.49	**98.66 ± 1.75**	98.18 ± 2.12	92.35 ± 3.39	97.17 ± 1.25	77.22 ± 5.24	83.37 ± 2.64	81.60 ± 12.93	92.41 ± 5.27
4	99.76 ± 0.21	99.84 ± 0.28	92.59 ± 2.33	97.75 ± 0.93	**99.93 ± 0.14**	95.47 ± 1.83	97.42 ± 0.45	96.57 ± 2.10	98.65 ± 1.13
5	98.94 ± 1.22	99.97 ± 0.04	92.27 ± 6.48	98.36 ± 0.49	92.38 ± 8.11	99.81 ± 0.15	98.68 ± 0.27	99.58 ± 0.56	99.54 ± 0.26
6	68.73 ± 5.93	68.01 ± 5.07	83.31 ± 1.79	68.69 ± 3.78	84.14 ± 3.04	**87.37 ± 5.13**	59.97 ± 2.85	66.69 ± 4.83	79.71 ± 7.61
OA	95.71 ± 0.53	96.22 ± 0.50	91.74 ± 1.80	93.88 ± 0.43	94.46 ± 3.29	95.56 ± 0.57	93.13 ± 0.23	93.77 ± 1.33	**96.63 ± 0.45**
AA	93.34 ± 0.92	93.79 ± 0.51	92.07 ± 0.80	90.52 ± 0.71	93.90 ± 2.14	90.96 ± 1.17	87.95 ± 0.74	88.68 ± 3.15	**94.12 ± 1.42**
Kappa	94.26 ± 0.71	94.93 ± 0.67	88.94 ± 2.39	91.82 ± 0.57	92.73 ± 4.18	94.06 ± 0.76	90.82 ± 0.31	91.68 ± 1.76	**95.49 ± 0.60**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Classification maps generated by different models on the disjointed UT dataset. (a) SSRN. (b) A2S2K-ResNet. (c) SS-ConvNeXt. (d) ViT. (e) SSTN. (f) SSFTT. (g) SpectralFormer. (h) MFT. (i) MSNAT. (j) Ground truth.

TABLE VI

COMPUTATIONAL COST COMPARISON OF DIFFERENT MODELS ON DIFFERENT DATASETS

Dataset	Computational Cost	Models								
		SSRN	A2S2K-ResNet	SS-ConvNeXt	ViT	SSTN	SSFTT	SpectralFormer	MFT	MSNAT
UP	Parameters	216,537	220,653	13,093,385	1,092,041	15,623	484,417	178,217	306,473	119,758
	Inference Time (s)	10.67	12.50	22.60	11.48	6.57	5.97	11.69	6.55	7.02
UH	Parameters	278,127	283,251	13,099,087	1,095,055	17,734	673,735	239,984	312,303	121,869
	Inference Time (s)	4.17	4.90s	7.10	3.74	1.99	1.94	4.53	2.07	2.39
UT	Parameters	155,022	158,178	13,089,286	1,089,286	13,612	299,902	137,582	262,758	117,747
	Inference Time (s)	4.42	5.22	15.13	6.98	2.59	2.25	5.09	2.42	3.37

TABLE VII

CLASSIFICATION RESULTS OF THE MSNAT WITH AND WITHOUT ST/MSFE. THE BEST RESULTS ARE SHOWN IN BOLD

Dataset	Module		Metrics		
	ST	MSFE	OA	AA	Kappa
UP	✓		83.17 ± 2.92	89.21 ± 1.18	78.44 ± 3.56
		✓	**93.39 ± 2.48**	**94.00 ± 1.12**	**91.24 ± 3.23**
	✓	✓	93.34 ± 1.02	92.16 ± 1.23	91.11 ± 1.36
UH	✓		82.56 ± 0.84	83.90 ± 0.74	81.10 ± 0.89
		✓	**87.31 ± 0.22**	**89.43 ± 0.12**	**86.29 ± 0.23**
	✓	✓	86.26 ± 0.47	88.62 ± 0.52	85.16 ± 0.50
UT	✓		94.49 ± 1.48	90.83 ± 1.62	92.64 ± 1.94
		✓	95.95 ± 0.36	93.44 ± 0.52	94.58 ± 0.48
	✓	✓	**96.63 ± 0.45**	**94.12 ± 1.42**	**95.49 ± 0.60**

TABLE VIII

COMPARISON BETWEEN PCM AND MLP

Datasets	Methods	Metrics		
		OA	AA	Kappa
UP	PCM	**93.34 ± 1.02**	92.16 ± 1.23	**91.11 ± 1.36**
	MLP	93.08 ± 0.91	**92.94 ± 1.11**	90.80 ± 1.20
UH	PCM	**86.26 ± 0.47**	**88.62 ± 0.52**	**85.16 ± 0.50**
	MLP	85.95 ± 0.75	88.26 ± 0.94	84.82 ± 0.82
UT	PCM	**96.63 ± 0.45**	**94.12 ± 1.42**	**95.49 ± 0.60**
	MLP	96.17 ± 0.45	93.44 ± 1.33	94.87 ± 0.61

4) Impact of Local Window Scales: Finally, the effect of multiscale features extracted by the MSNA module is investigated. Specifically, the original MSNA module is modified by removing either the upper or lower branch to obtain models with different local window sizes. The model with a local window size of 3 only retains the upper branch with small scale, while the model with a local window size of 5 only keeps the lower branch with a larger scale. These modified models are evaluated on the UP, UH, and UT datasets, and their classification results are compared with the original MSNAT model incorporating multiscale features. Table IX illustrates the corresponding classification results which indicate that the proposed MSNAT model with multiscale features achieves better classification performance on all the three datasets than the modified models with single-scale features, demonstrating the importance of incorporating multiscale information for HSI classification.

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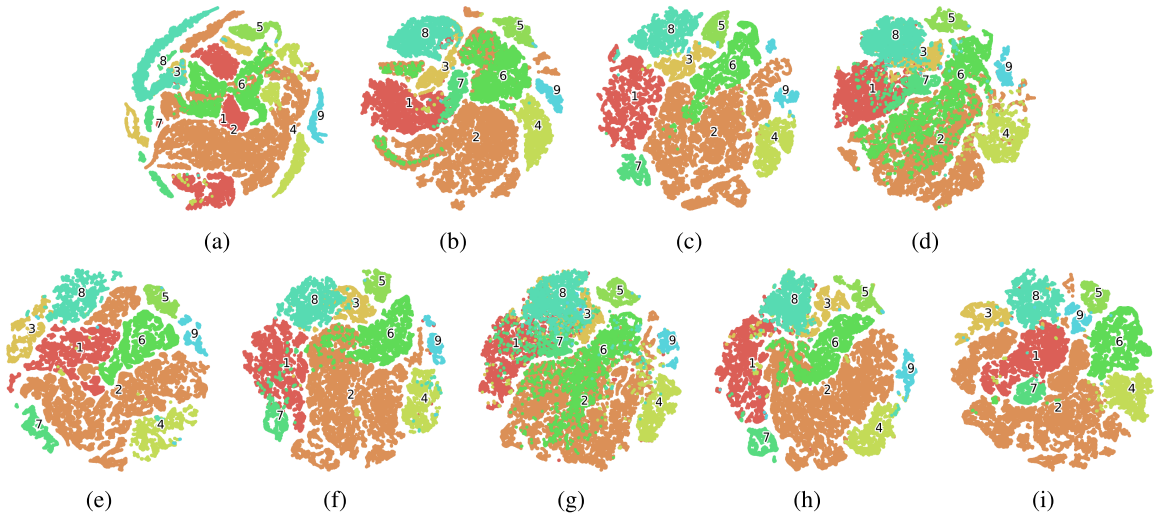


Fig. 9. t-SNE visualization of extracted features for the UP dataset. (a) SSRN. (b) A2S2K-ResNet. (c) SS-ConvNeXt. (d) ViT. (e) SSTN. (f) SSFTT. (g) SpectralFormer. (h) MFT. (i) MSNAT.

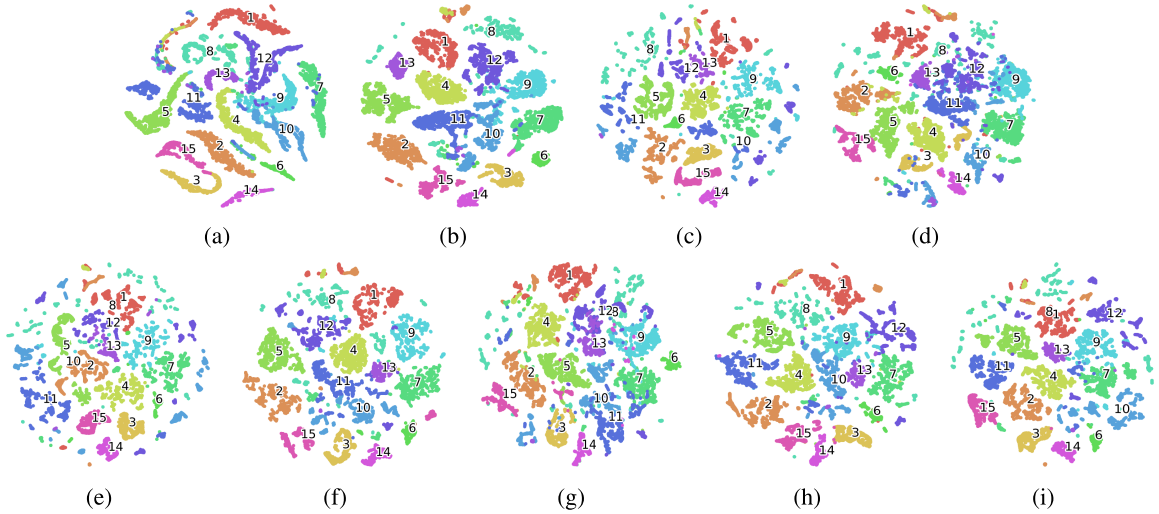


Fig. 10. t-SNE visualization of extracted features for the UH dataset. (a) SSRN. (b) A2S2K-ResNet. (c) SS-ConvNeXt. (d) ViT. (e) SSTN. (f) SSFTT. (g) SpectralFormer. (h) MFT. (i) MSNAT.

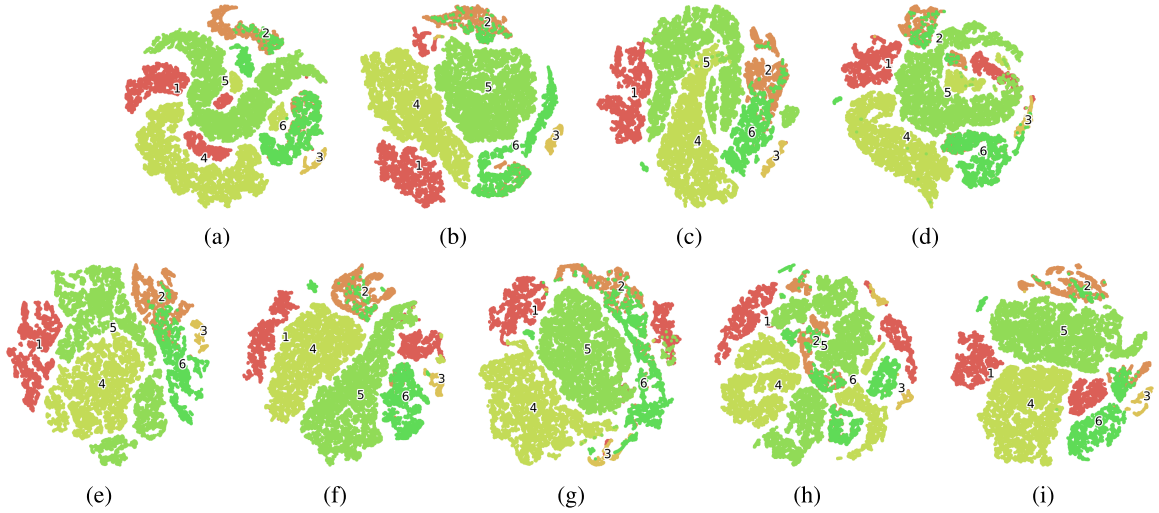


Fig. 11. t-SNE visualization of extracted features for the UT dataset. (a) SSRN. (b) A2S2K-ResNet. (c) SS-ConvNeXt. (d) ViT. (e) SSTN. (f) SSFTT. (g) SpectralFormer. (h) MFT. (i) MSNAT.

It is also observed that for the UT and UH datasets, the model with smaller local window size performs better than that with larger window size. This can be attributed to the influence of the spatial resolution and class distribution of the datasets.

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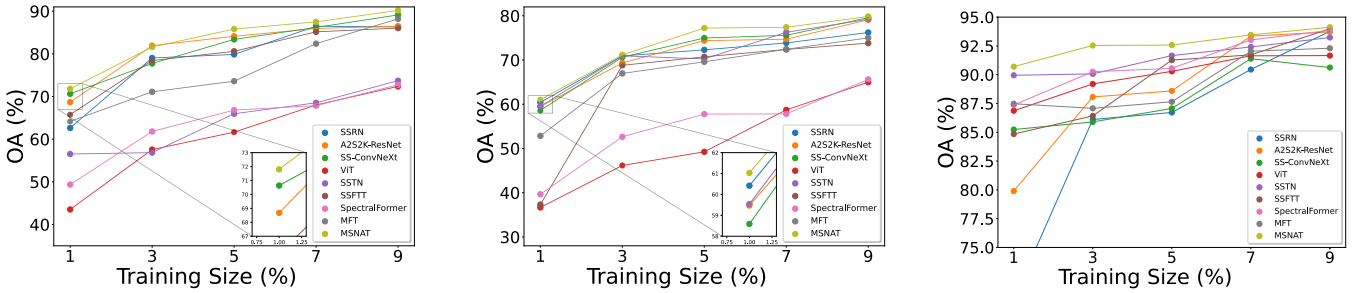


Fig. 12. Classification results over different sizes of training data. (a) UP. (b) UH. (c) UT.

TABLE IX
CLASSIFICATION RESULTS OF MSNAT WITH DIFFERENT LOCAL WINDOW SCALES. THE BEST RESULTS ARE SHOWN IN BOLD

Dataset	Scales	Metrics		
		OA	AA	Kappa
UP	3	91.42 ± 2.14	93.02 ± 0.68	88.73 ± 2.71
	5	92.06 ± 1.26	92.41 ± 0.54	89.56 ± 1.61
	3, 5	**93.34 ± 1.02**	92.16 ± 1.23	**91.11 ± 1.36**
	7, 9	93.08 ± 0.92	**93.25 ± 0.90**	90.79 ± 1.22
UH	3	85.74 ± 0.67	88.04 ± 0.63	84.60 ± 0.72
	5	85.50 ± 0.71	87.74 ± 0.48	84.34 ± 0.77
	3, 5	**86.26 ± 0.47**	**88.62 ± 0.52**	**85.16 ± 0.50**
	7, 9	85.61 ± 0.59	88.11 ± 0.59	84.45 ± 0.64
UT	3	96.24 ± 0.99	93.72 ± 1.80	94.95 ± 1.33
	5	95.94 ± 0.37	93.97 ± 1.06	94.60 ± 0.49
	3, 5	**96.63 ± 0.45**	**94.12 ± 1.42**	**95.49 ± 0.60**
	7, 9	96.08 ± 0.68	93.99 ± 1.27	94.75 ± 0.91

For the UT dataset which has a higher spatial resolution, smaller window sizes may capture more fine-grained details, leading to better classification performance. For the UH dataset which has more complex class distribution, a smaller window size may better capture local features, while a larger window size may oversmooth the class boundaries and result in a lower classification accuracy.

With the aim of extracting local multiscale features and considering the prevalent utilization of kernel sizes in CNNs, scales 3 and 5 are selected in the MSFE module. To further explore the impact with different scales, experiments with larger scales 7 and 9 are undertaken. The classification accuracies across the three datasets are presented in Table IX. It could be observed that the results with larger scales 7 and 9 are slightly lower than the results with scales 3 and 5. This phenomenon could be attributed to the fact that larger scales could incorporate more pixels belonging to different classes while they offer more spatial information, thus reducing the accuracy.

IV. CONCLUSION

In this article, a novel MSNAT network with optimized spatial pattern is proposed for HSIs' classification. Multiscale spatial features which are often ignored in most existing transformer-based models are fully explored in this study. These features are effectively extracted and optimized through the designed MSFE and ST modules, respectively. The proposed MSNAT method is validated on three real-world HSI datasets, and the results show notable advantages. First, the comparison between models with single- and multiscale features demonstrates that multiscale features are crucial for accurate HSI classification. For instance, the incorporation of multiscale features leads to more than 1% improvement in overall accuracy on the UP and UH datasets. It is also found that the proposed method exhibits a strong capability for multiscale features' extraction, which significantly enhances the classification performance, achieving state-of-the-art overall accuracy of 93.34%, 86.26%, and 96.63% on the UP, UH, and UT datasets, respectively. Besides, the proposed MSNAT method is proven to be robust when limited training samples are available. This is evidenced by the fact that when small subsets with 3% of the original training samples are used for training, the overall accuracy on the UT dataset decreases from 96.63% to 92.51%. It can also be concluded that the proposed method is suitable for real-world applications where lack of labeled data is a common challenge. Furthermore, the ST module is illustrated to be effective for handling the spatial variability in HSIs through image transformation, including rotation and translation, further improving the performance of the proposed MSNAT method.

It should also be noted that the ST module in the MSNAT network is influenced by the characteristics of the HSI, such as its spatial resolution and class distribution. Therefore, future research will be devoted to improving the generalization of the ST module in diverse scenarios.

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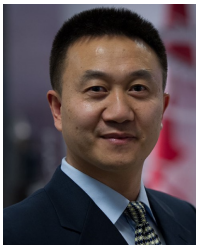
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